Pablo Miralles Ph.D. Candidate

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Education

Technical University of Madrid — Madrid, Spain
Ph.D. in Artificial Intelligence — 2024 - Now
- Advisor: Prof. Alejandro Martín
- Research area: Deep Learning for Natural Language Processing

Technical University of Madrid — Madrid, Spain
M.Sc. in Machine Learning and Big Data — 2023 - 2024
- GPA: 9.65/10.00, Rank: 1st

University of Murcia — Murcia, Spain
B.Sc. in Mathematics — 2017 - 2022
- GPA: 9.74/10.00, Rank: 1st

University of Murcia — Murcia, Spain
B.Sc. in Computer Science — 2017 - 2022
- GPA: 9.57/10.00, Rank: 2nd

Publications

1. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2026). Evaluating the scalability and adversarial generalization of GRPO-trained NLI models. *Knowledge-Based Systems*, 352, 117013. https://doi.org/10.1016/j.knosys.2026.117013
2. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2026). Not all tokens are created equal: Perplexity Attention Weighted Networks for AI-generated text detection. *Information Fusion*, 125, 103465. https://doi.org/10.1016/j.inffus.2025.103465
3. Girón, A., **Miralles-González, P.**, Huertas-Tato, J., D'Antonio, S., & Camacho, D. (2026). xList-Hate: A Checklist-Based Framework for Interpretable and Generalizable Hate Speech Detection. *arXiv*. https://doi.org/10.48550/arXiv.2602.05874
4. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2025). LLM one-shot style transfer for Authorship Attribution and Verification. *arXiv*. https://doi.org/10.48550/arXiv.2510.13302
5. **Miralles-González, P.**, Huertas-Tato, J., Martín, A., & Camacho, D. (2025). On the locality bias and results in the Long Range Arena. *arXiv*. https://doi.org/10.48550/arXiv.2501.14850

Industry Experience

Fever | Software Engineer — 2023.02 - 2023.08
- Backend: Python, Flask and Django.
- Lead and developed a project to automatically structure information and generate plan content using LLMs, massively reducing the costs.

Orenes | Software Engineer Intern — 2021.07 - 2021.08
- Backend: C# and ASP.NET. Frontend: ReactNative.
- Kickstarted the development of the mobile app.

Projects and grants

FPU Fellowship (University Teacher Training Programme)
Spanish Ministry of Science and Innovation 2025.01 - Now

- Highly competitive predoctoral contract aimed at training future university lecturers and researchers.

XAI-DisInfodemics: eXplainable AI for disinformation and conspiracy detection during infodemics
European Union NextGenerationEU/PRTR 2024.05 - 2025.01

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FightDIS: Fighting against Information DISorders in Online Social Networks
Spanish Ministry of Science and Innovation 2023.09 - 2024.05

Awards and Honors

- **Best academic record**, M.Sc. in Machine Learning and Big Data 2024
- **Extraordinary End of Degree Award**, BSc. in Mathematics 2023
- **Extraordinary End of Degree Award**, BSc. in Computer Science 2023
- **Academic Excellence Award in joint BSc. programs**, BSc. in Math+CS 2023
- **23th/97**, Southwestern Europe Regional Contest (SWERC) 2021-2022, 2022
- **17th/109**, Southwestern Europe Regional Contest (SWERC) 2020-2021, 2021
- **National Collaboration Grant**, Spanish Ministry of Science and Innovation 2021
- **18th/98**, Southwestern Europe Regional Contest (SWERC) 2019-2020, 2020

Skills

Languages: Spanish (native), English (C1).
Programming: Python, C++, Java.
Deep Learning: PyTorch, HuggingFace, vLLM, Triton (basic), CUDA (basic).
Data Science/Machine Learning: SKLearn, Numpy, Pandas.
Experiment management/distributed: WanDB, RAY.
Other: Docker, Linux, Git, Latex.

Teaching

Data Structures | Computer Science BSc (Asst. Prof.) 2025/26 — present
- Teaching core data structures including queues, stacks, lists, trees, graphs, and hash tables.

Classical Prediction Methods | Data Science BSc (Asst. Prof.) 2025/26
- Lectures and practical sessions on linear regression, clustering algorithms, and time series models.

Generative Models | Master in Deep Learning 2024/25 — present
- Teaching generative modeling: VAEs, GANs, autoregressive models, and diffusion models.

Deep Learning for NLP | Master in Deep Learning 2024/25 — present
- Instruction on real-world NLP applications using HuggingFace, LangGraph, vLLM, and vector databases.

Algorithms and Data Structures | Data Science BSc (Asst. Prof.) 2024/25
- Taught graph theory fundamentals and greedy algorithm design.